

1 Summary

Model (Run-name): model1 (test)

Iteration: 19

TAC: 154847.25933978206

Took: 2.1518211364746094s

Total Time: 29.079179048538208s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

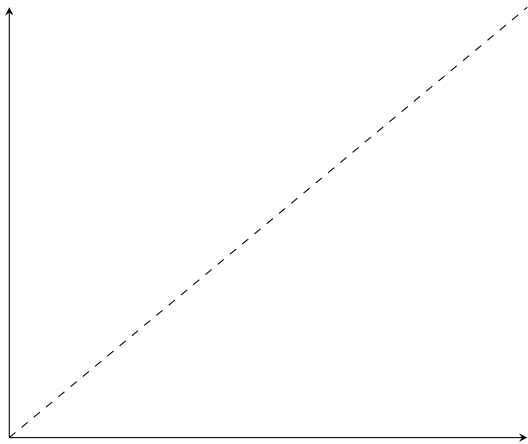
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

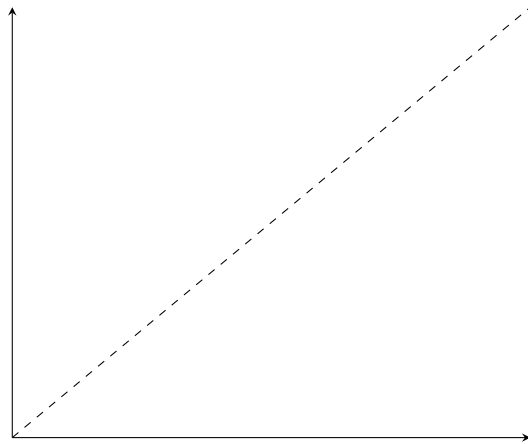
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.05244145711752263	$b_{1,1,1}^{H,out}$	889.6216028171018
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017464023096765917	$b_{1,2,2}^{H,out}$	3876.772841111382
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.029709249761038215	$b_{2,1,2}^{H,out}$	4103.904212803715
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.01452807871202422	$b_{1,1,1}^{C,in}$	5990.588947865486
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009599207330710935	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.02259563987620934	$b_{2,1,2}^{C,in}$	4211.120302410845
$A_{1,1,1}^\beta$	70.46247894104941	$q_{1,1,1}$	673.2271588886176	$b_{1,1,1}^{C,out}$	6663.816106754104
$A_{1,2,2}^\beta$	68.10969007738701	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	144.67005099128556	$q_{2,1,2}$	2440.159261667073	$b_{2,1,2}^{C,out}$	6651.2795640779195
$A_{cu,1}^\beta$	5.081618324442152	$q_{cu,1}$	176.77284111138238		
$A_{cu,2}^\beta$	37.612770977560544	$q_{cu,2}$	1959.8407383329268		
$A_{hu,1}^\beta$	13.189852638760229	$q_{hu,1}$	486.6135794443088		
$A_{1,1,1}$	70.46247894104941	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	68.10969007738701	$t_{1,2}^{(H)}$	582.6772841111382		
$A_{2,1,2}$	144.67005099128556	$t_{1,3}^{(H)}$	387.67728411113825		
$A_{cu,1}$	5.081618324442152	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.612770977560544	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.189852638760229	$t_{2,3}^{(H)}$	467.9920369166463		
$dt_{1,1,1}$	32.44090529628724	$t_{1,1}^{(C)}$	617.5590947037127		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	572.6772841111382		
$dt_{1,2,2}$	82.67728411113823	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	37.677284111138235	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	17.32271588886176	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	57.9920369166463	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	67.67728411113823	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	147.99203691664633	$t_{1,2,2}^H$	387.6772841111382		
$dt_{hu,1}$	62.440905296287255	$t_{2,1,2}^H$	370.0		
$f_{1,1,1}^H$	2.4043827103164914	$t_{1,1,1}^C$	636.9614739994557		
$f_{1,2,2}^H$	10.0	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	11.09163300757761	$t_{2,1,2}^C$	647.0579928591765		
$f_{1,1,1}^C$	10.461882513729234	$b_{1,1,1}^{H,in}$	1562.8487617057194		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5826.772841111382		
$f_{2,1,2}^C$	10.27102512783133	$b_{2,1,2}^{H,in}$	6544.0634744707895		

3 *RecLMTD* Tangent Points

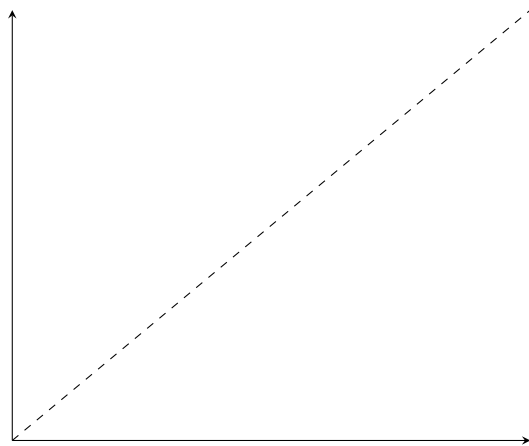
- $(1, 1, 1)$
 - Value: 0.052441
 - Error: 0.000000056587768
 - Relative error: 0.000001079064359
- $(1, 2, 2)$
 - Value: 0.017464
 - Error: 0.000000143940952
 - Relative error: 0.000008242073693
- $(2, 1, 2)$
 - Value: 0.029709
 - Error: 0.000000786666590
 - Relative error: 0.000026478142893
- $cu, 1$
 - Value: 0.014528
 - Error: 0.000000024402050
 - Relative error: 0.000001679644638
- $cu, 2$
 - Value: 0.009599
 - Error: 0.000000019144809
 - Relative error: 0.000001994411614
- $hu, 1$
 - Value: 0.022596
 - Error: 0.000000006140586
 - Relative error: 0.000000271759691



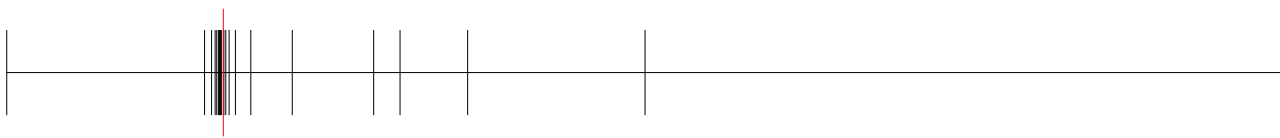
(a) *RecLMTD* - (1, 1, 1)



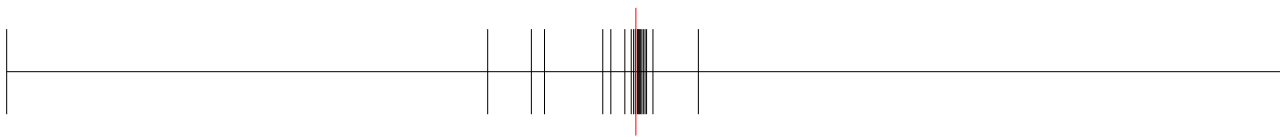
(b) *RecLMTD* - (1, 2, 2)



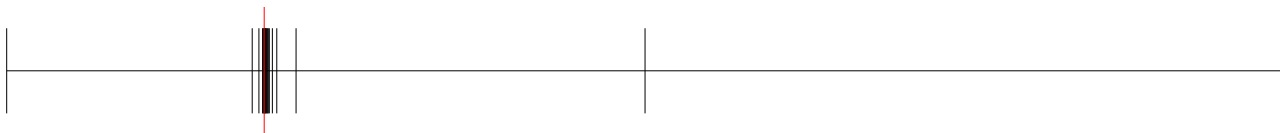
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - cu, 1



(b) *RecLMTD* - cu, 2



(a) *RecLMTD* - hu, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1562.8487617057194
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5826.772841111382
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6544.0634744707895
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 889.6216028171018
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3876.772841111382
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 4103.904212803715
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 5990.588947865486
 - Absolute error: 0.6935167867786731
 - Relative error: -0.00011575431318258252
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4211.120302410845
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 6663.816106754104
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 6651.2795640779195
 - Absolute error: 5.330660257212912
 - Relative error: 0.0008020916703329385



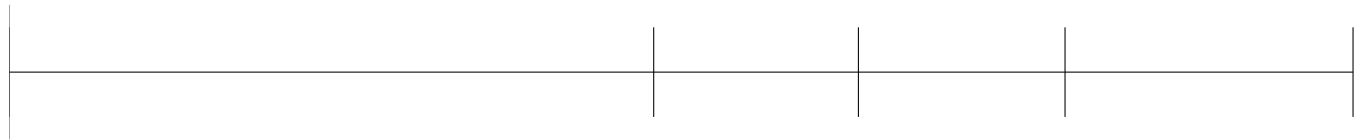
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



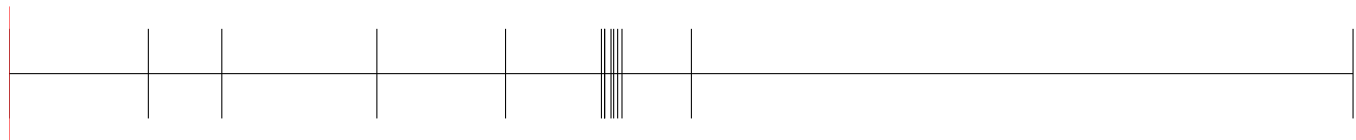
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



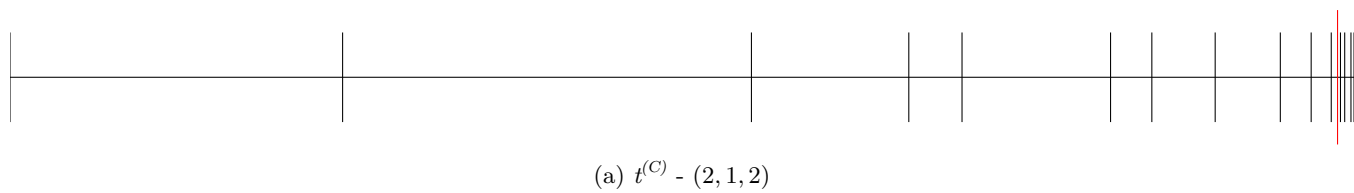
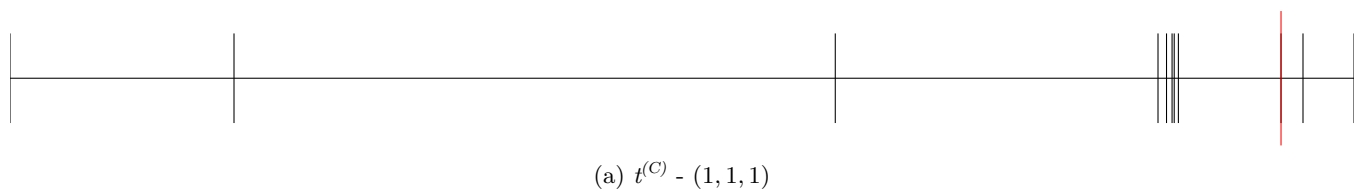
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

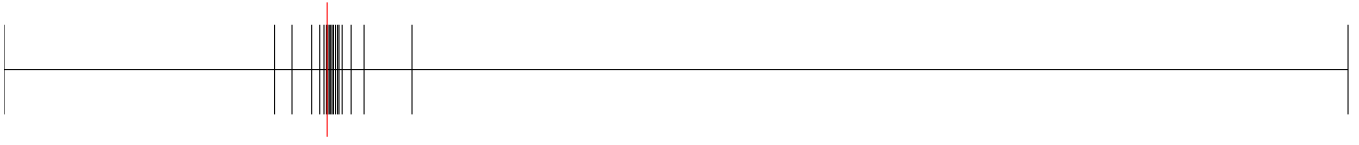
- $(1, 1, 1)$
 - Value: 70.462479
 - Error: 0.147547425368657
 - Relative error: 0.002089610115750
- $(1, 2, 2)$
 - Value: 68.109690
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 144.670051
 - Error: 0.320550931869803
 - Relative error: 0.002210839375918

5.2 A_{CU}

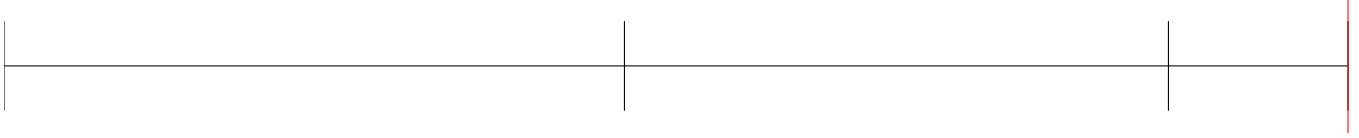
- $cu, 1$
 - Value: 5.081618
 - Error: 0.054721175186477
 - Relative error: 0.010653730188675
- $cu, 2$
 - Value: 37.612771
 - Error: 0.013064187302177
 - Relative error: 0.000347213217858

5.3 A_{HU}

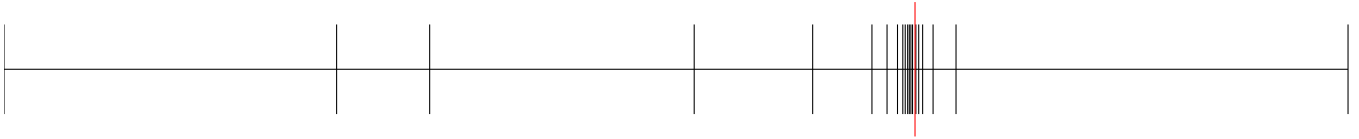
- $hu, 1$
 - Value: 13.189853
 - Error: 0.004561601235912
 - Relative error: 0.000345722148247



(a) $A^\beta - (1, 1, 1)$



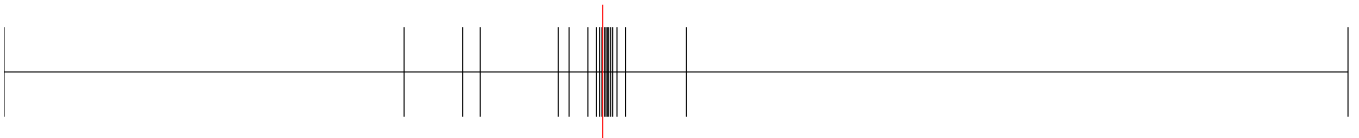
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

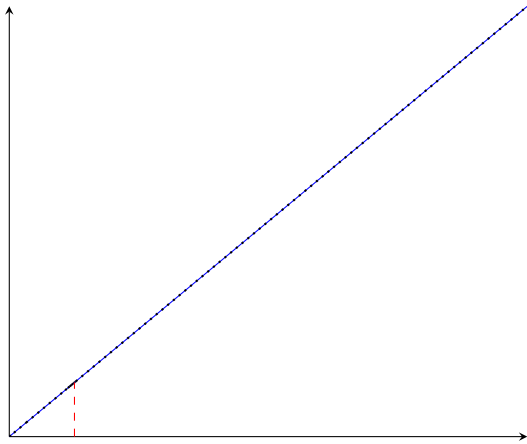


(a) $A^\beta - hu, 1$

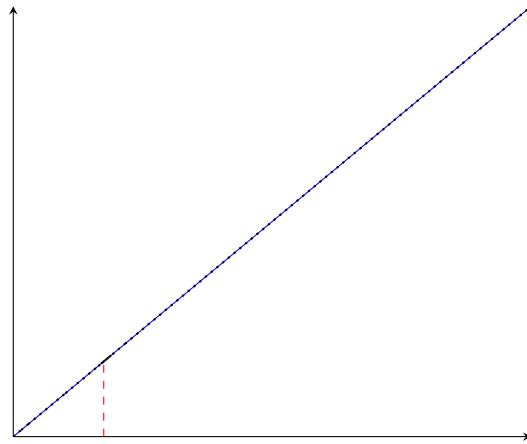
6 A^β Breakpoints

- (1, 1, 1)
 - q : 70.462479
 - $\hat{q}^\beta$: 70.462479
 - q^β : 70.462479
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, **70.30181758862284**, **70.50425945122109**, 70.60541450524272, 70.95870595008942, 71.22933662920877, 71.2616536557244, 71.6637831223933, 71.88123217857378, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 68.109690
 - $\hat{q}^\beta$: 68.109690
 - q^β : 68.109690
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, **67.91344539650363**, **68.66676327863097**, 69.0314457108091, 69.34502840170047, 69.41355730879985, 70.37078617087542, 70.94081287188642, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 144.670051
 - $\hat{q}^\beta$: 144.670051
 - q^β : 144.670051
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, 132.12228195956155, 135.81963163113545, 139.47623233772285, 140.7008934199032, **142.4338700260237**, **144.70843093515643**, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 5.081618
 - $\hat{q}^\beta$: 5.081618
 - q^β : 5.081618
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, 4.432280503220532, 4.613047202566896, 4.825601156120121, 4.849719301830286, 4.974201802143407, 4.983938626310074, 4.993068227518925, **5.0418124100090695**, **7.702824927089129**, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000

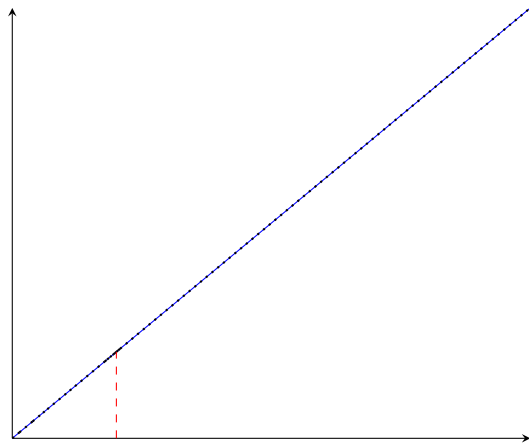
- $cu, 2$
 - q : 37.612771
 - $\hat{q}^\beta$: 37.612771
 - q^β : 37.612771
 - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, **37.554741031910254**, **37.6551299822246**, 37.682669717644174, 37.78573239590169, 37.9018730666985, 37.9454585315849, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.189853
 - $\hat{q}^\beta$: 13.189853
 - q^β : 13.189853
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, **13.161020122653593**, **13.192252946941991**, 13.213749911872785, 13.218222920629291, 13.22717920580809, 13.244985069124171, 13.269097009439808, 13.30871858348944, 13.32663103908035, 13.346967780023082, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



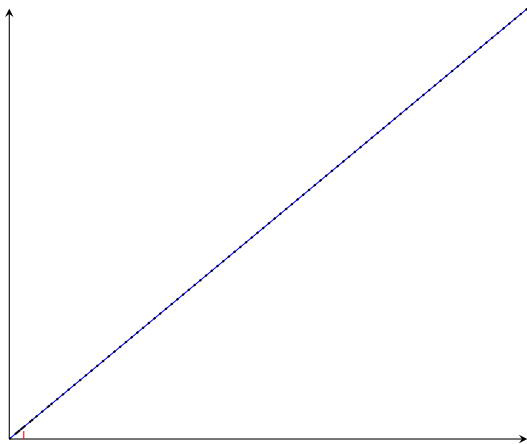
(a) $q^\beta - (1, 1, 1)$



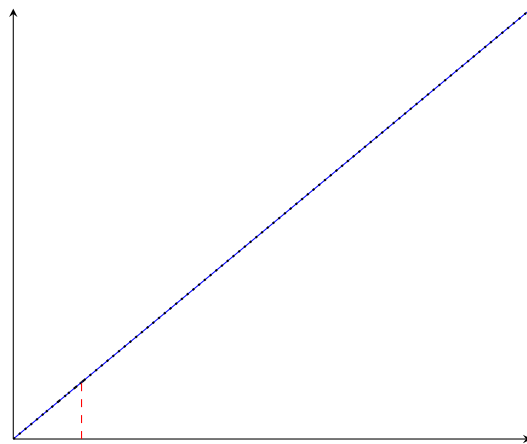
(b) $q^\beta - (1, 2, 2)$



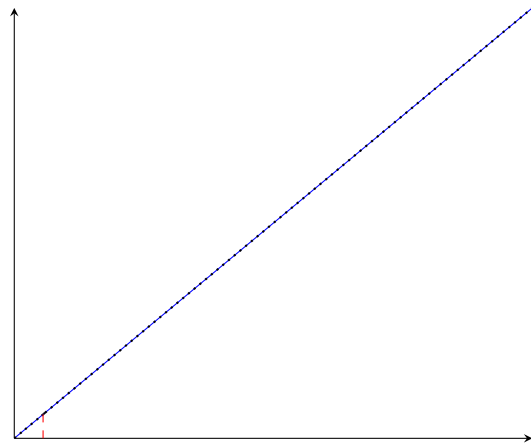
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	13.0	$b_{2,2,2}^{H,out}$	5255.9365255292105
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1938.879697589155
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	0.0
$z_{2,2,1}$	-0.0	$RecLMTD_{1,2,1}$	0.005370550548886958	$b_{2,1,1}^{C,in}$	2599.570313801587
$z_{2,2,2}$	0.0	$RecLMTD_{2,1,1}$	0.1	$b_{2,2,1}^{C,in}$	6500.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.006079999999999996	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.03507702960436037	$b_{1,1,2}^{C,out}$	1938.8796975891541
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	0.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	2599.570313801587
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	6500.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	370.0		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	10.0	$t_{2,2,2}^H$	590.0		
$dt_{1,2,1}$	280.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	10.0	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	220.0	$t_{2,1,1}^C$	537.3839353112696		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	500.0		
$dt_{2,2,3}$	113.84994470322357	$t_{2,2,2}^C$	370.04544034995354		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	0.0		
$f_{1,1,2}^H$	0.0	$b_{1,2,1}^{H,in}$	4937.15123829428		
$f_{1,2,1}^H$	7.595617289683508	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	5255.9365255292105		
$f_{2,2,2}^H$	8.90836699242239	$b_{1,1,2}^{H,out}$	0.0		
$f_{1,1,2}^C$	4.728974872168671	$b_{1,2,1}^{H,out}$	4937.15123829428		
$f_{1,2,1}^C$	0.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	4.538117486270766	$b_{2,2,1}^{H,out}$	0.0		